Supplementary Information

Table I. Trade-off evaluation between predictive performance (Classification/Segmentation) and hardware-aware metrics for TFLite-exported for the Tomato crop.

Model	**Classification**			**Segmentation**		**#Epoch**	**Total Loss**	**Weight (TFLite) [MB]**	**Parameters [M]**	**Inference [ms]**
	Accuracy [%]	**Recall [%]**	**F1-score [%]**	**Mean IoU [%]**	**Accuracy [%]**					
U-Net & ResNet 50	94.64	94.64	94.64	40.05	90.18	27	0.5039	18.11	32.37	298.00
U-Net & MobileNetV2	95.83	95.83	95.78	16.27	81.36	48	0.4797	11.94	3.15	151.07
Mask R-CNN & ResNet50	55.36	55.36	48.11	32.10	81.86	6	3.2282	117.14	30.80	94.40
Mask R-CNN & MobileNetV2	92.86	92.86	92.75	54.90	87.27	7	2.6227	24.84	6.55	67.41
DeepLabV3 & ResNet50	94.64	94.64	94.53	67.70	91.11	12	0.5793	119.97	31.55	146.32
DeepLabV3 & MobileNetV2	98.21	98.21	98.21	66.03	90.20	22	0.4461	20.79	5.49	87.98

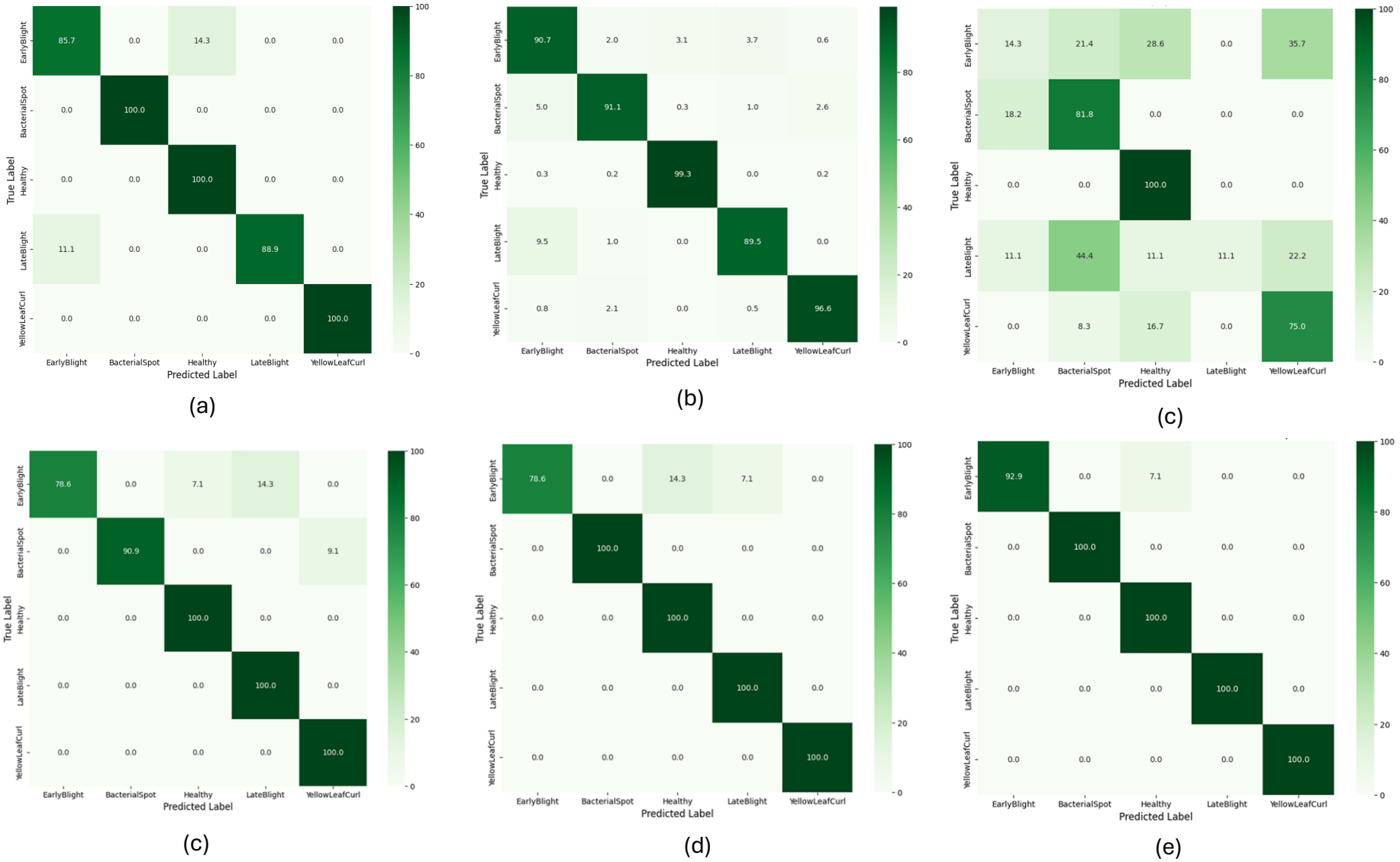


Fig. 1. Confusion matrix model of models in Tomato crop, (a) U-Net and ResNet50 (b) U-Net and MobilNetV2 (c) Mask R-CNN and ResNet50 (d) Mask R-CNN and MobilNetV2 (e) DeepLabV3+and ResNet50 (f) DeepLabV3+ and MobileNetV2

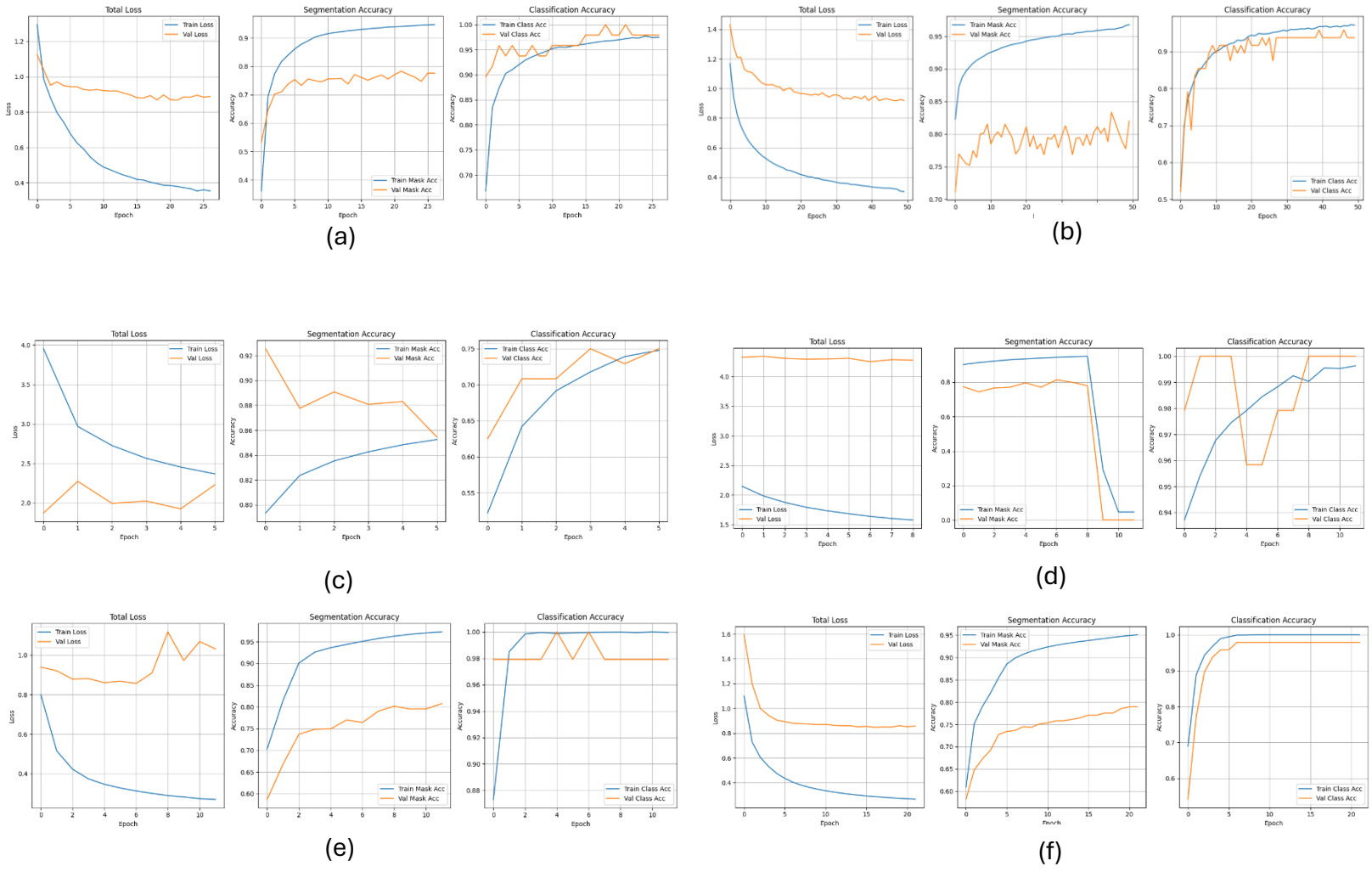


Fig.2. Loss curves in Tomato crop, (a) U-Net and ResNet50 (b) U-Net and MobilNetV2 (c) Mask R-CNN and ResNet50 (d) Mask R-CNN and MobilNetV2 (e) DeepLabV3+and ResNet50 (f) DeepLabV3+ and MobileNetV2

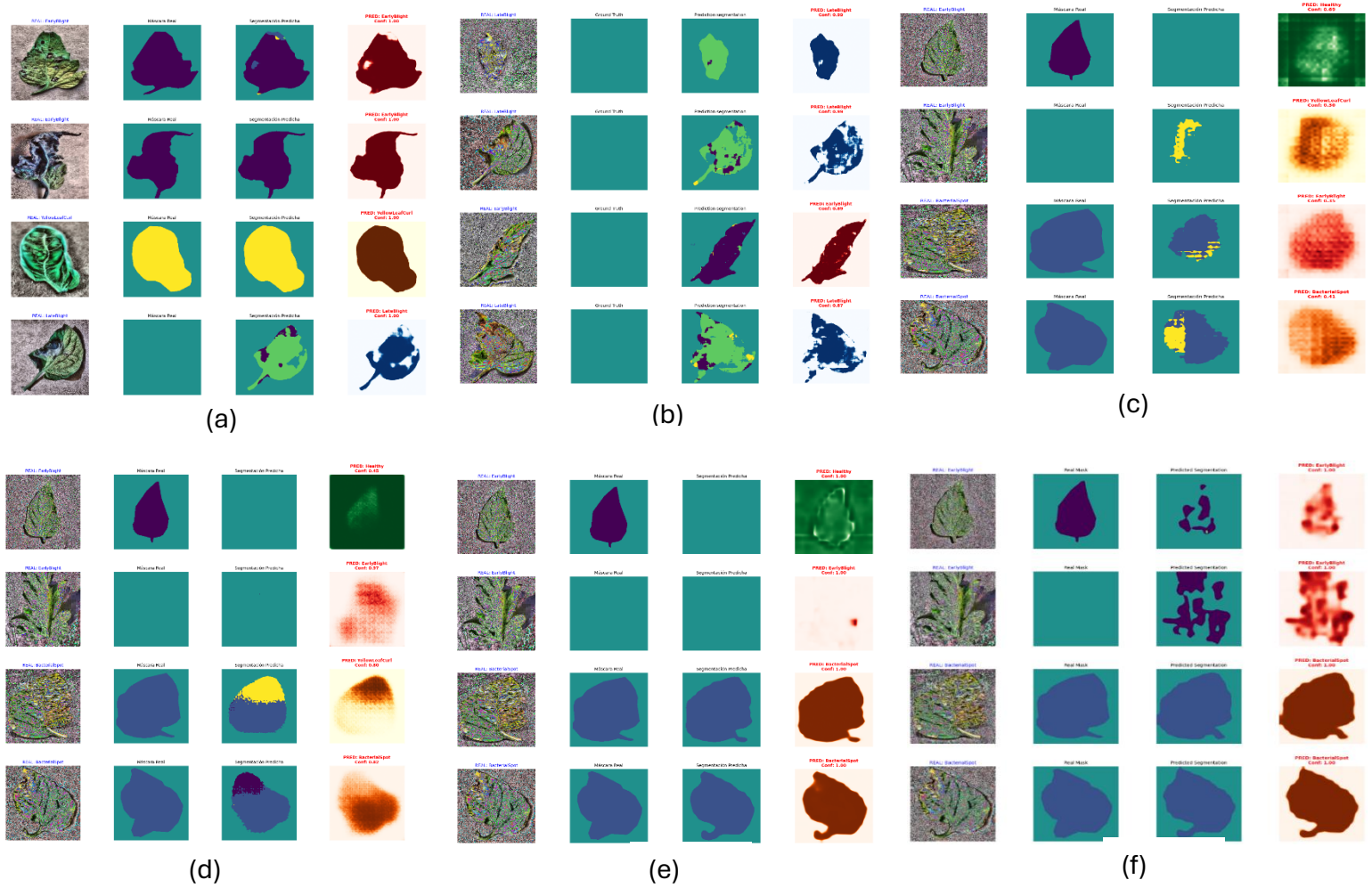


Fig. 3. Segmentation Samples for Tomato Crops with Corresponding Masks. (a) U-Net and ResNet50 (b) U-Net and MobilNetV2 (c) Mask R-CNN and ResNet50 (d) Mask R-CNN and MobilNetV2 (e) DeepLabV3+and ResNet50 (f) DeepLabV3+ and MobileNetV2

Table III. Trade-off evaluation between predictive performance (Classification/Segmentation) and hardware-aware metrics for TFLite-exported for Potato crop

Model	**Classification**			**Segmentation**		**#Epoch**	**Total Loss**	**Weigth (TFLite) [MB]**	**Parameters [M]**	**Inference [ms]**
	Accuracy [%]	**Recall [%]**	**F1-score [%]**	**Mean IoU [%]**	**Accuracy [%]**					
U-Net & ResNet50	96.43	96.43	96.27	34.27	76.76	21	0.6115	119.86	31.52	214.40
U-Net& MobileNetV2	98.21	98.21	98.19	81.89	91.33	13	0.2255	13.1	3.32	68.72
Mask R-CNN & ResNet50	83.93	83.93	83.56	49.25	76.43	13	0.5983	102.75	27.03	188.03
Mask R-CNN & MobileNetV2	96.43	96.43	96.27	35.61	71.22	14	0.7926	22.37	5.91	84.99
DeepLabV3+ & ResNet50	94.64	94.64	94.57	58.90	80.09	11	0.4641	110.65	29.10	514.75
DeepLabV3+ & MobileNetV2	98.21	98.21	98.19	42.73	44.35	14	0.9978	18.11	4.79	123.02

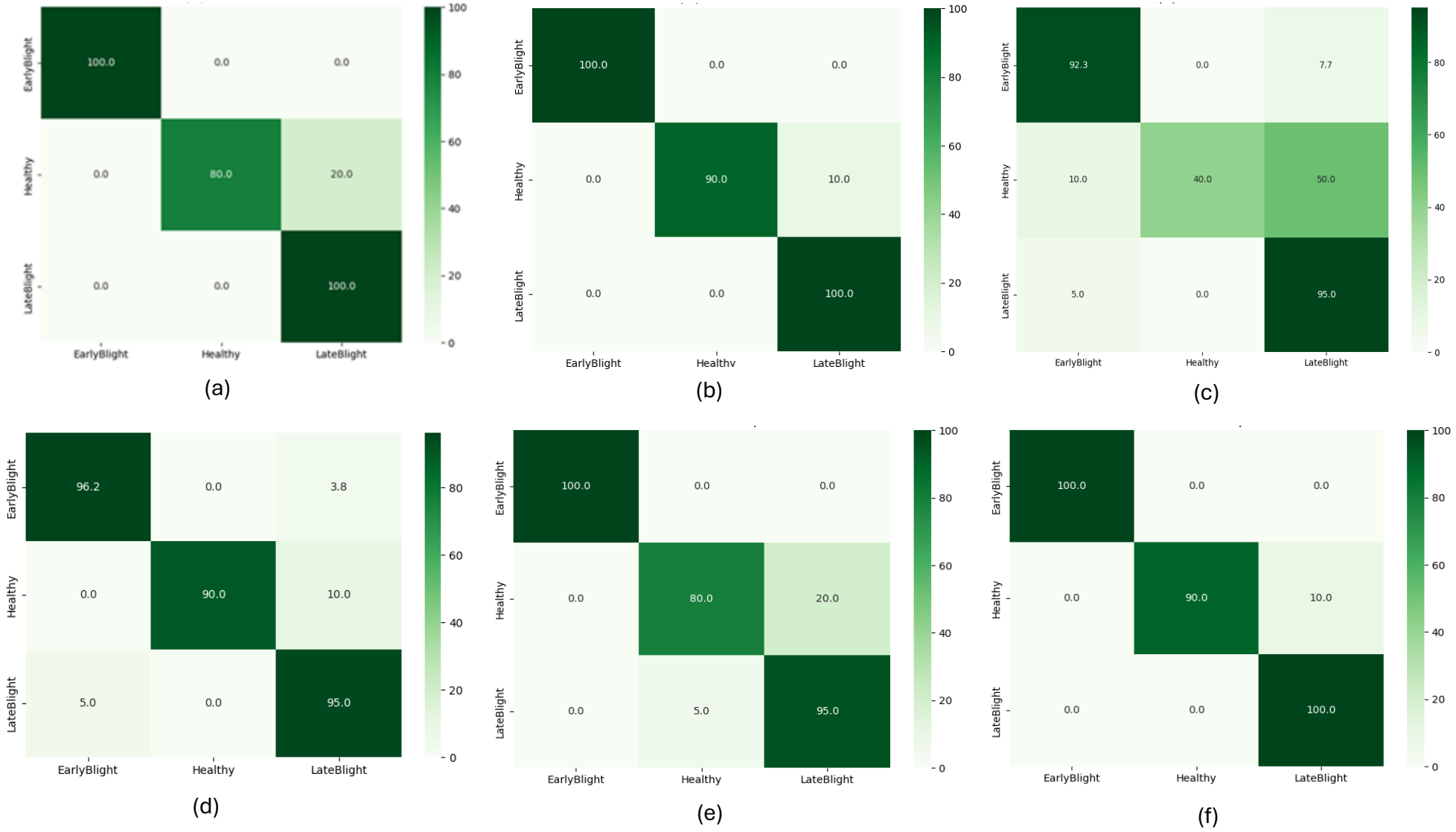


Fig. 4. Confusion matrix model of models in Potato crop, (a) U-Net and ResNet50 (b) U-Net and MobilNetV2 (c) Mask R-CNN and ResNet50 (d) Mask R-CNN and MobilNetV2 (e) DeepLabV3+and ResNet50 (f) DeepLabV3+ and MobileNetV2.

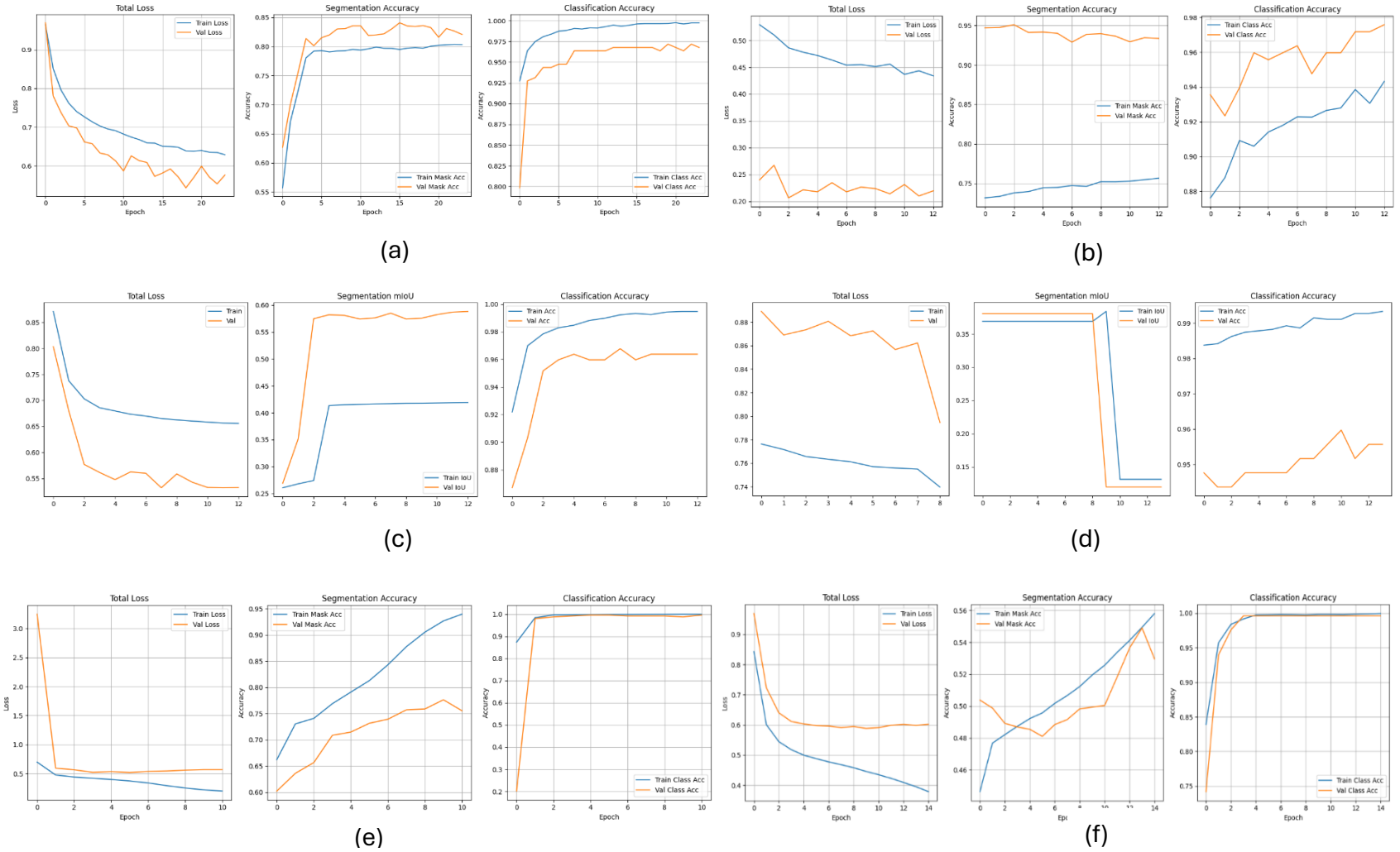


Fig.5. Loss curves in Potato crop, (a) U-Net and ResNet50 (b) U-Net and MobilNetV2 (c) Mask R-CNN and ResNet50 (d) Mask R-CNN and MobilNetV2 (e) DeepLabV3+and ResNet50 (f) DeepLabV3+ and MobileNetV2

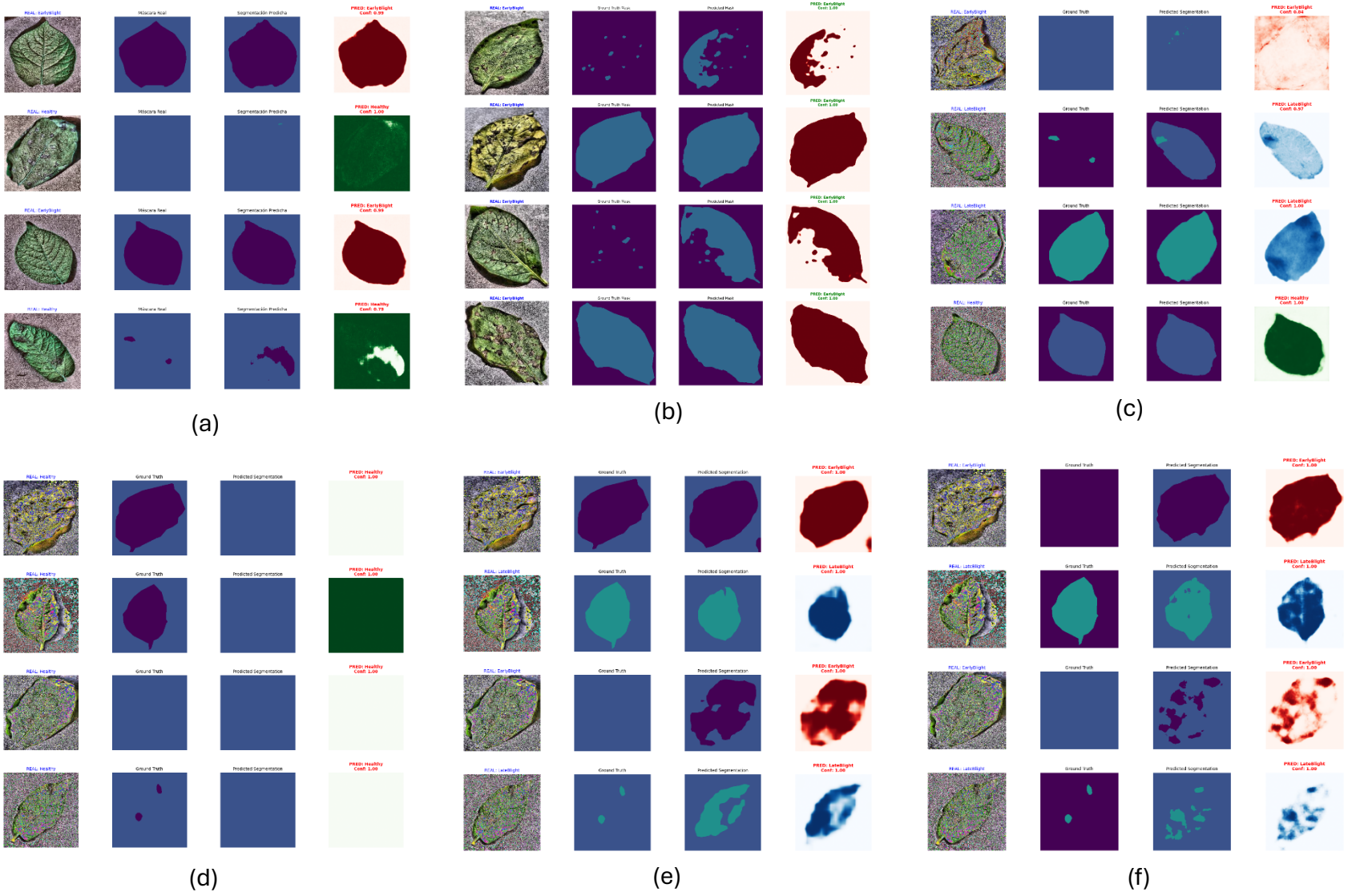


Fig. 6. Segmentation Samples for Potato Crops with Corresponding Masks. (a) U-Net and ResNet50 (b) U-Net and MobilNetV2 (c) Mask R-CNN and ResNet50 (d) Mask R-CNN and MobilNetV2 (e) DeepLabV3+and ResNet50 (f) DeepLabV3+ and MobileNetV2

Table 3. Trade-off evaluation between predictive performance (Classification/Segmentation) and hardware-aware metrics for TFLite-exported for Bell Pepper crop

Model	Classification			Segmentation		#Epoch	Total Loss	Weigth (TFLite) [MB]	Parameters [M]	Inference [ms]
	Accuracy [%]	Recall [%]	F1-score [%]	Mean IoU [%]	Accuracy [%]					
U-Net & ResNet50	100	100	100	97.87	99.14	16	0.0213	5.05	1.34	130.52
U-Net& MobileNetV2	91.67	91.67	91.55	97.04	98.73	13	0.0562	13.38	3.56	87.05
Mask R-CNN & ResNet50	100	100	100	98.27	99.33	11	0.0293	91.18	23.94	105.65
Mask R-CNN & MobileNetV2	100	100	100	96.59	98.42	16	0.0408	11.23	32.76	87.65
DeepLabV3 & ResNet50	100	100	100	98.71	99.50	9	0.0357	40.33	10.59	141.61
DeepLabV3 & MobileNetV2	100	100	100	98.91	99.63	38	0.0234	9.88	2.63	59.81

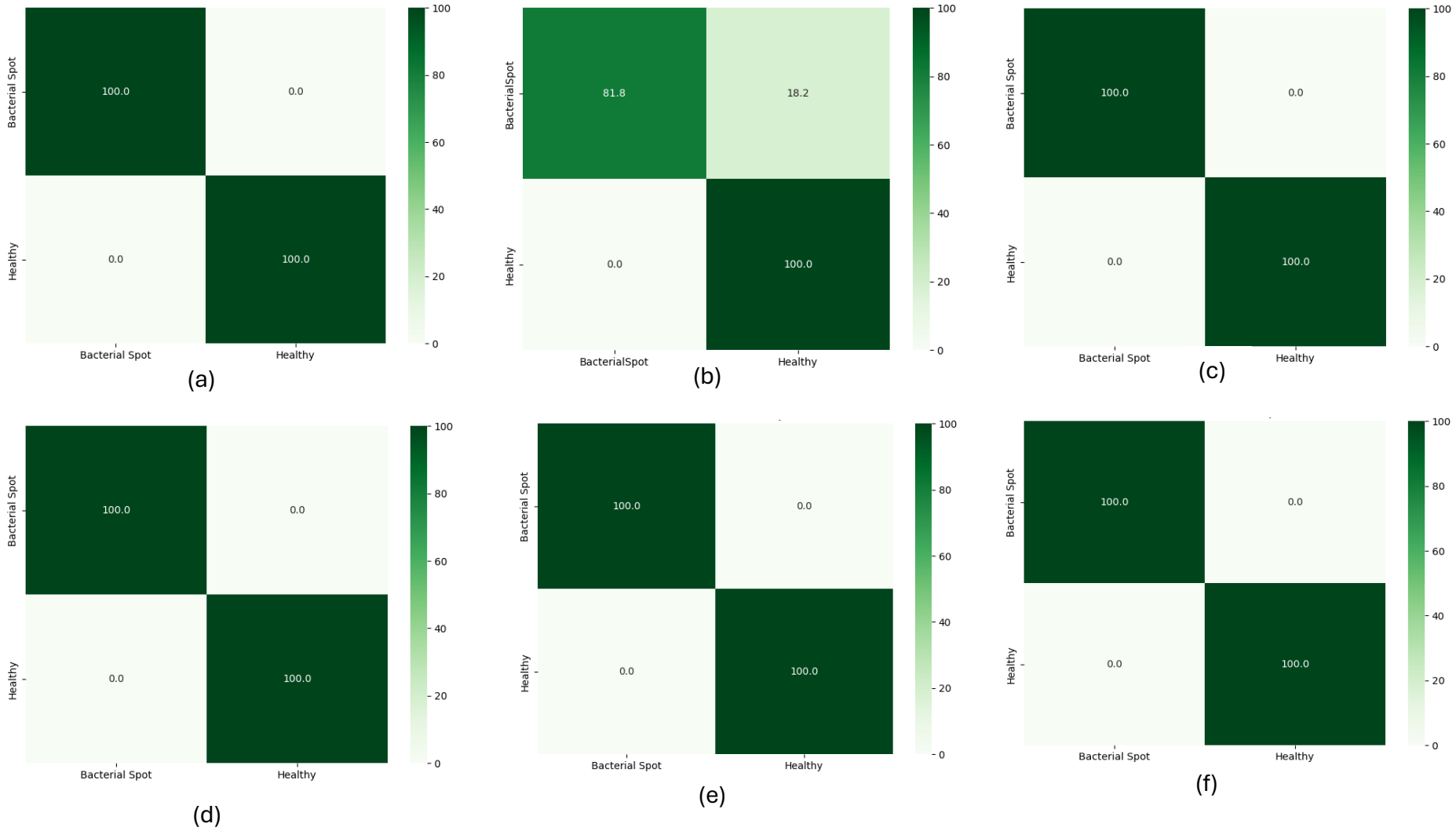


Fig. 7. Confusion matrix model of models in Bell Pepper crop, (a) U-Net and ResNet50 (b) U-Net and MobilNetV2 (c) Mask R-CNN and ResNet50 (d) Mask R-CNN and MobilNetV2 (e) DeepLabV3+and ResNet50 (f) DeepLabV3+ and MobileNetV2

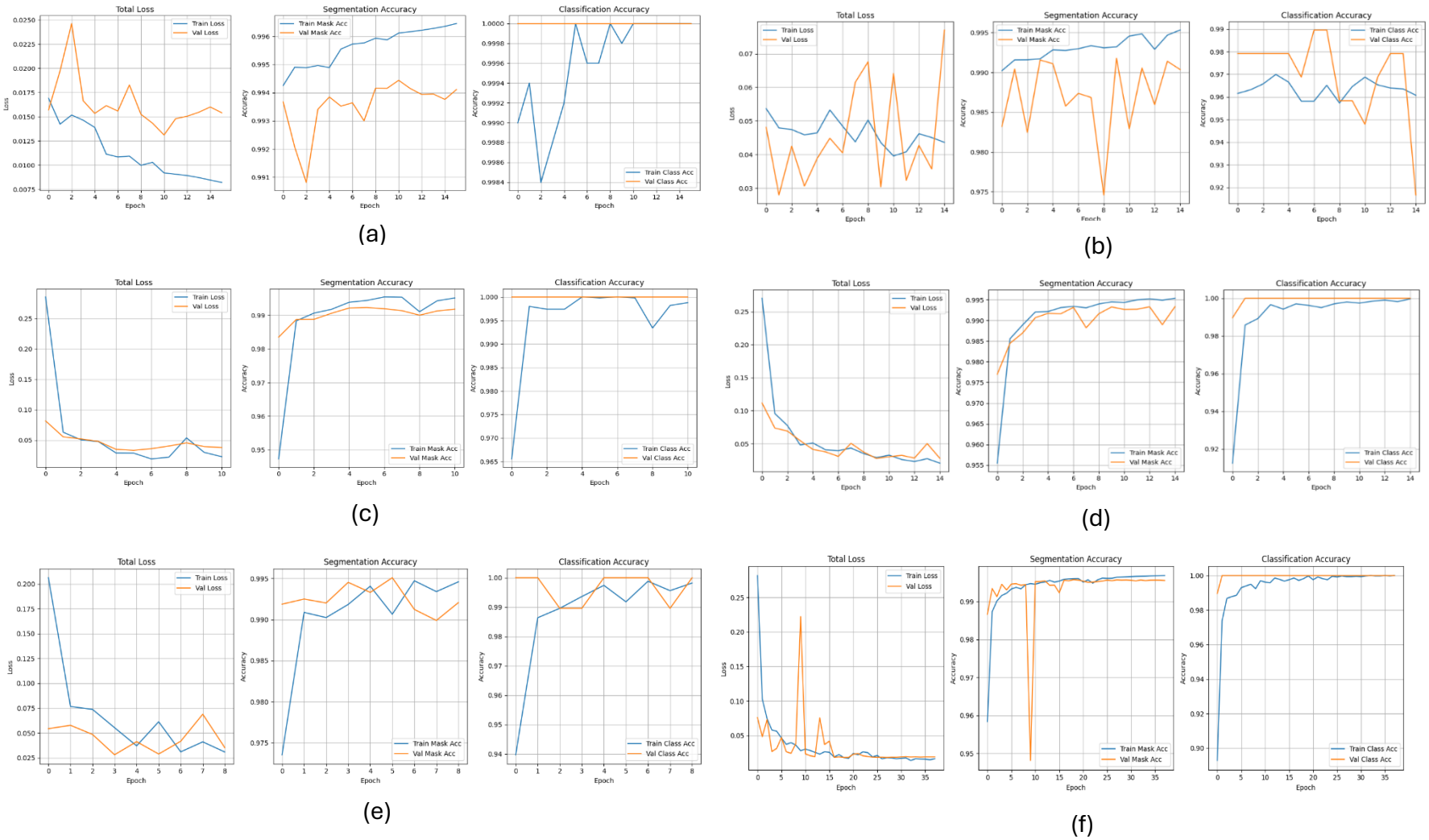


Fig.8. Loss curves in Bell Pepper crop, (a) U-Net and ResNet50 (b) U-Net and MobilNetV2 (c) Mask R-CNN and ResNet50 (d) Mask R-CNN and MobilNetV2 (e) DeepLabV3+and ResNet50 (f) DeepLabV3+ and MobileNetV2

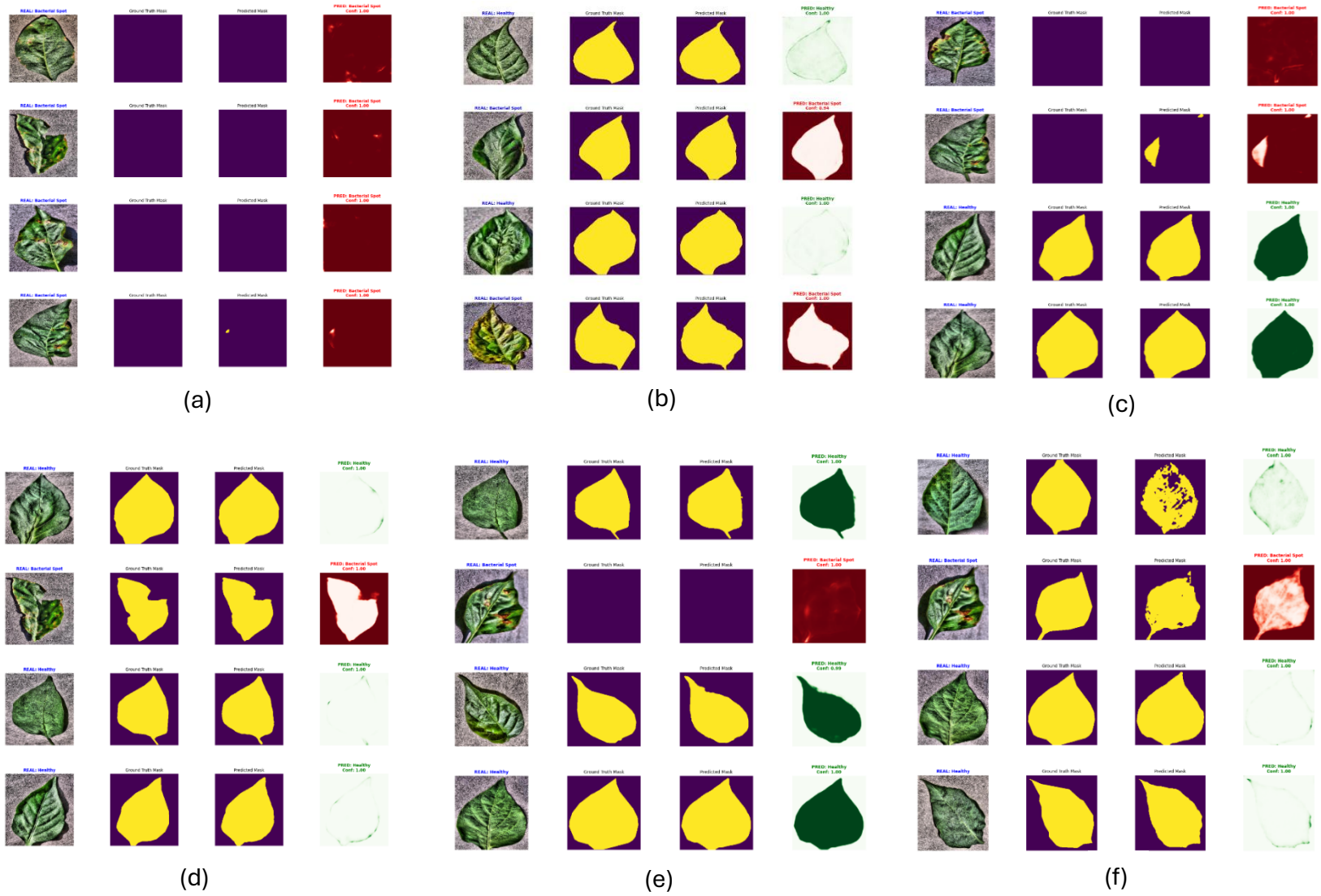


Fig. 9. Segmentation Samples for Bell Pepper Crops with Corresponding Masks. (a) U-Net and ResNet50 (b) U-Net and MobilNetV2 (c) Mask R-CNN and ResNet50 (d) Mask R-CNN and MobilNetV2 (e) DeepLabV3+and ResNet50 (f) DeepLabV3+ and MobileNetV2

Table 4. Trade-off evaluation between predictive performance (Classification/Segmentation) and hardware-aware metrics for TFLite-exported for Eggplant crop

Model	Classification			Segmentation		#Epoch	Total Loss	Weigth (TFLite) [MB]	Parameters [M]	Inference [ms]
	Accuracy [%]	Recall [%]	F1-score [%]	Mean IoU [%]	Accuracy [%]					
U-Net & ResNet 50	97.50	100	100	82.11	94.33	23	1.1344	167.46	43.99	267.65
U-Net& MobileNetv2	97.50	95	94.96	78.52	91.89	33	1.5599	13.41	3.55	77.73
Mask R-CNN & ResNet50	97.50	97.50	97.49	78.99	93.01	13	1.1926	123.98	32.59	128.57
Mask R-CNN & MobileNetv2	97.50	97.50	97.49	75.53	90.17	18	1.1097	12.62	3.35	66.59
DeepLabV3 & ResNet50	97.50	97.50	97.49	77.55	92.23	10	1.1479	115.65	30.41	137.12
DeepLabV3 & MobileNetv2	97.50	97.50	97.49	81.69	93.51	21	0.5465	9.89	2.63	66.99

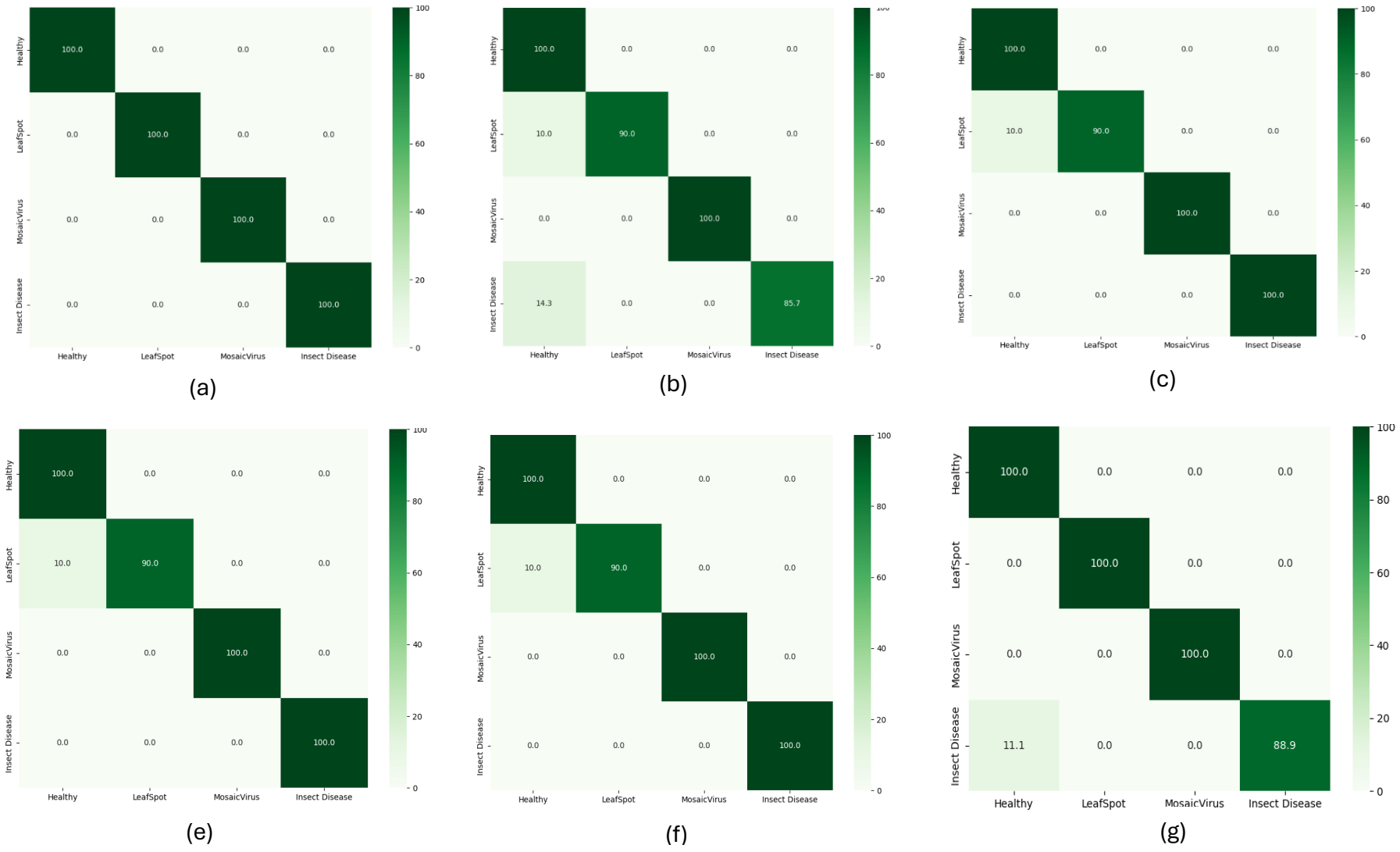


Fig. 10. Confusion matrix model of models in Eggplant crop. (a) U-Net and ResNet50 (b) U-Net and MobilNetV2 (c) Mask R-CNN and ResNet50 (d) Mask R-CNN and MobilNetV2 (e) DeepLabV3+and ResNet50 (f) DeepLabV3+ and MobileNetV2.

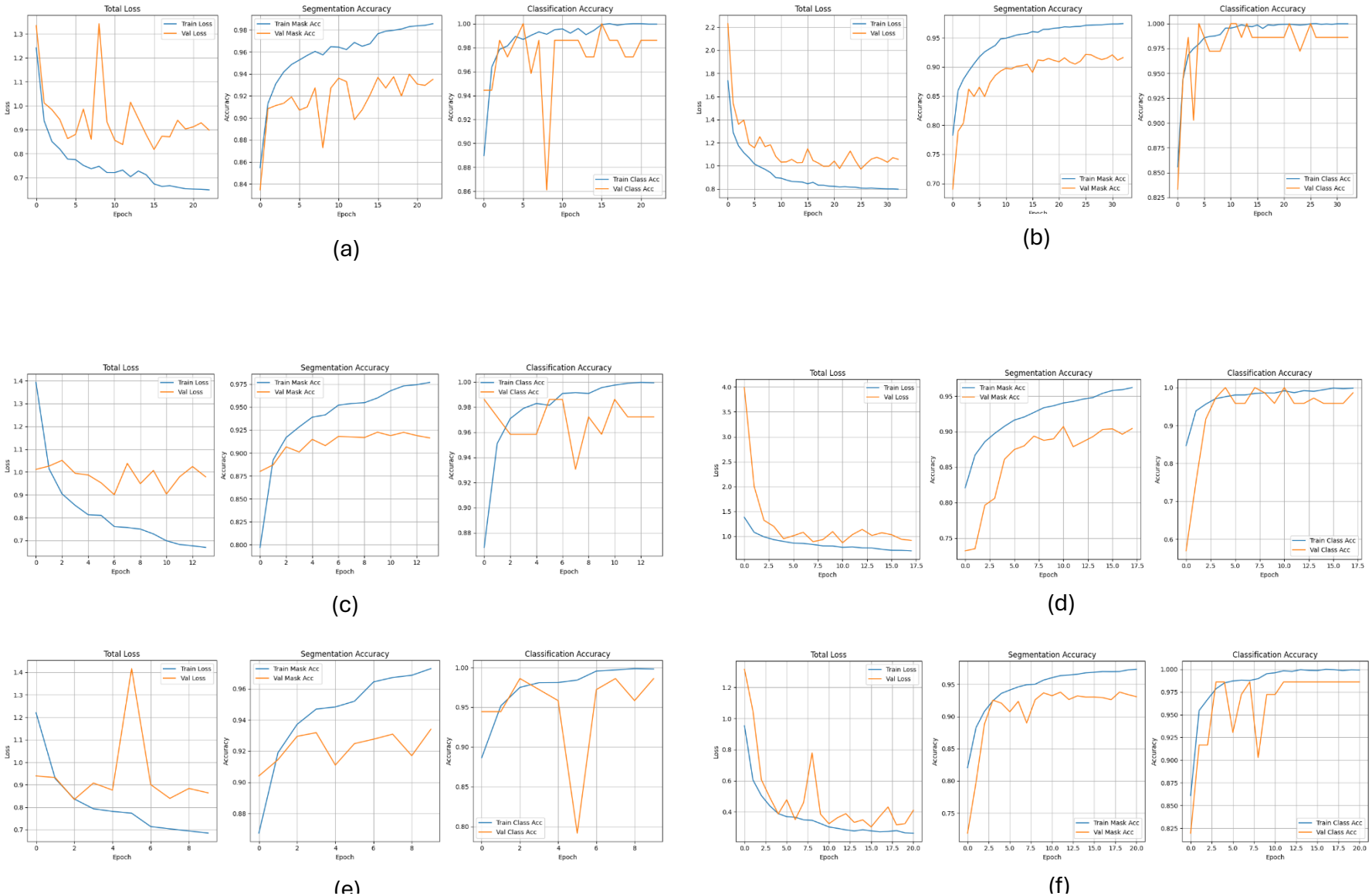


Fig.11. Loss curves in Eggplant crop, (a) U-Net and ResNet50 (b) U-Net and MobilNetV2 (c) Mask R-CNN and ResNet50 (d) Mask R-CNN and MobilNetV2 (e) DeepLabV3+and ResNet50 (f) DeepLabV3+ and MobileNetV2.

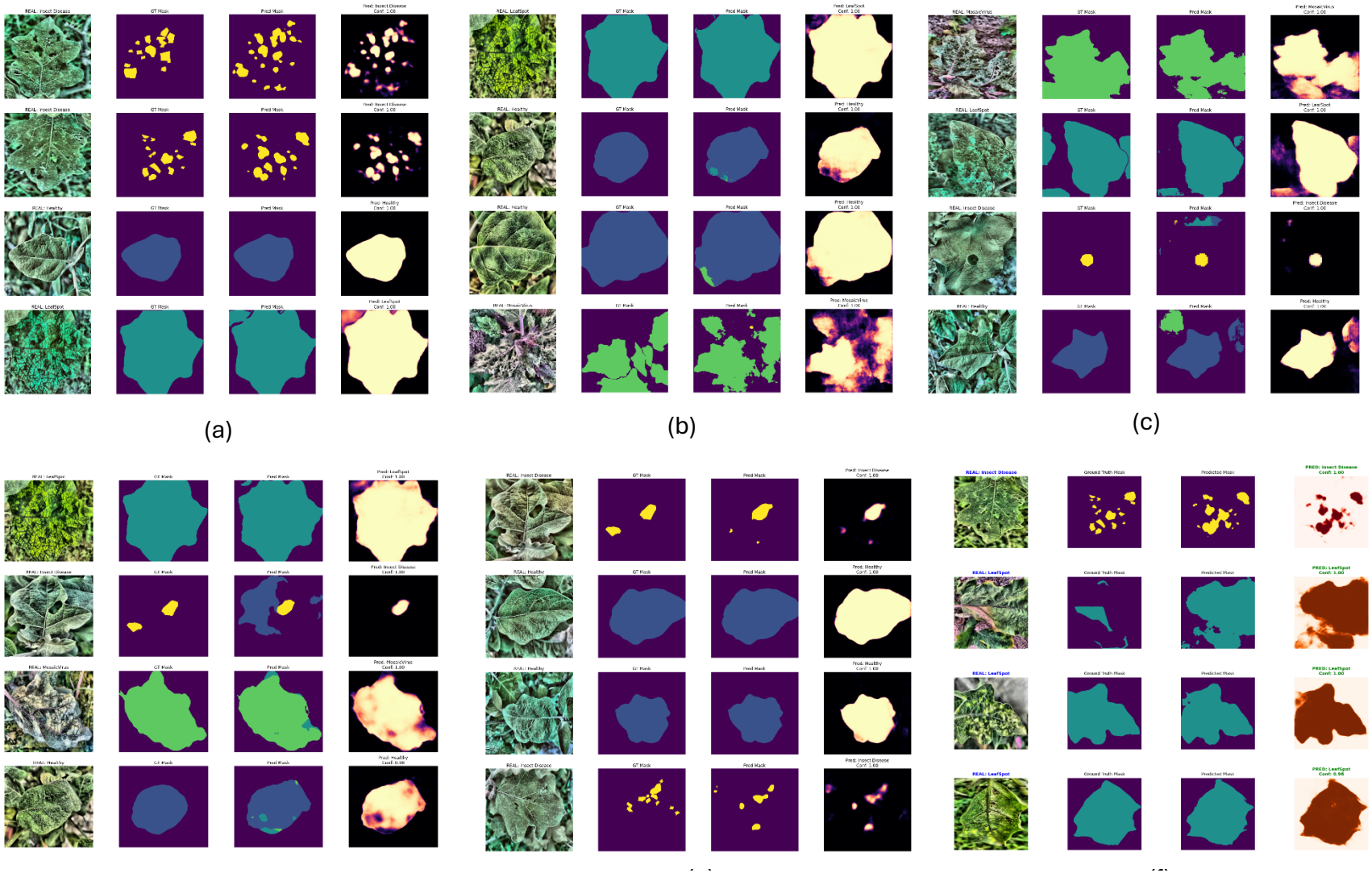


Fig. 12. Segmentation Samples for Eggplant Crops with Corresponding Masks. (a) U-Net and ResNet50 (b) U-Net and MobilNetV2 (c) Mask R-CNN and ResNet50 (d) Mask R-CNN and MobilNetV2 (e) DeepLabV3+and ResNet50 (f) DeepLabV3+ and MobileNetV2

Survey S1. User Satisfaction Survey

User Satisfaction Survey: SolanAPP Project

Institution: National Center of Animal and Plant Health (CENSA) / Agrarian University of Havana (UNAH)

Date: _____ / _____ / 2026

Participant Profile: [ ] Farmer, [ ] Agronomy Student/Scholar, [ ] Agronomist/Field Technician

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Instructions:

Please rate your experience with SolanAPP by marking the box that best represents your opinion for each statement, where:

- **1:** Strongly Disagree
- **2:** Disagree
- **3:** Neutral
- **4:** Agree
- **5:** Strongly Agree

Section 1: Usability and Interface

#	Statement	1	2	3	4	5
1.1	The application interface is intuitive and easy to navigate.	[ ]	[ ]	[ ]	[ ]	[ ]
1.2	The visual icons and menus are clear and easy to understand.	[ ]	[ ]	[ ]	[ ]	[ ]
1.3	I felt confident using the app without constant technical assistance.	[ ]	[ ]	[ ]	[ ]	[ ]

Section 2: Functional Suitability and Transparency

#	Statement	1	2	3	4	5
2.1	Outlining the leaf spots in the picture helps me trust the diagnosis.	[ ]	[ ]	[ ]	[ ]	[ ]
2.2	The identification of the disease matches my visual observation.	[ ]	[ ]	[ ]	[ ]	[ ]
2.3	The diagnostic results provided by the app are accurate	[ ]	[ ]	[ ]	[ ]	[ ]

Section 3: LLM & Educational Value

#	Statement	1	2	3	4	5
3.1	The LLM-generated report successfully integrates both the computer vision diagnostic results and the user's qualitative field notes.	[ ]	[ ]	[ ]	[ ]	[ ]
3.2	The risk assessment provided by the LLM accurately correlates the disease's physical morphological features with the estimated percentage of affected area.	[ ]	[ ]	[ ]	[ ]	[ ]
3.3	The tailored action plans are clear and appropriately prioritize organic solutions for low/initial-severity levels over specific chemical treatments for critical cases.	[ ]	[ ]	[ ]	[ ]	[ ]
3.4	The practical management recommendations are useful and highly applicable to my daily field work or studies.	[ ]	[ ]	[ ]	[ ]	[ ]
3.5	The app helps me learn more about plant pathology while using it.	[ ]	[ ]	[ ]	[ ]	[ ]

Section 4: Reliability and Field Autonomy

#	Statement	1	2	3	4	5
4.1	The ability to perform offline diagnosis is essential for my work environment.	[ ]	[ ]	[ ]	[ ]	[ ]
4.2	I would use this app even if I have no internet access in the field.	[ ]	[ ]	[ ]	[ ]	[ ]
4.3	Sharing disease alerts by location (early warning) can help nearby farmers and the community	[ ]	[ ]	[ ]	[ ]	[ ]

Section 5: Qualitative Feedback (Optional)

What is the most useful feature of SolanAPP for you?

What improvements would you suggest for future versions?

Thank you for your participation.